



ADSSU Laboratory Asset Monitoring System (LAMS) User Manual

This manual serves as a guide for installing, accessing, navigating, and using the ADSSU Laboratory Asset Monitoring System (LAMS).



Agusan del Sur State University
Bunawan, Agusan del Sur



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BSIT 2C



June 2026



Agusan del Sur State University

• Bunawan, Agusan del Sur •

USER MANUAL

ADSSU LABORATORY ASSET MONITORING SYSTEM (LAMS)

ITCC 104 – RELATIONAL DATABASE MANAGEMENT SYSTEM



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Course / Year / Section:

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USER MANUAL INFORMATION

This user manual provides step-by-step instructions for installing, accessing, navigating, operating, maintaining, and troubleshooting the ADSSU Laboratory Asset Monitoring System (LAMS). It is intended for borrowers, administrators, and technical maintainers of the system.

Item	Description
System Title	ADSSU Laboratory Asset Monitoring System (LAMS)
Subject	ITCC 104 - Relational Database Management System
Manual Format	A4 landscape, book fold, printed back-to-back, staple wire binding at the center
Screenshot Requirement	Screenshot placeholders must be replaced with actual screenshots taken from the developed system before submission.
Primary Users	Administrator and Borrower/User
Prepared By	CM G. DAJAY - BSIT 2C

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I. SYSTEM INTRODUCTION

A. System Name

ADSSU Laboratory Asset Monitoring System (LAMS)

B. System Description

The ADSSU Laboratory Asset Monitoring System (LAMS) is a web-based information system designed to manage and monitor the borrowing and returning of laboratory assets and equipment at Agusan del Sur State University (ADSSU). The system enables authorized users or borrowers to browse available laboratory assets, submit borrow requests, track currently borrowed items, and view their complete borrow history. Administrators can review and approve or deny borrow requests, confirm returned items, and monitor overdue assets.

Key functionalities include:

- User registration and authentication
- Asset browsing with search and category filters
- Borrow request submission with quantity, return date, and purpose
- Approval workflow from pending to approved/denied/returned
- Overdue item detection and tracking
- Notification system for status updates
- Print and export capabilities for records
- Profile and password management

C. Intended Users

User Type	Role Description
Administrator	Reviews and approves/denies borrow requests; confirms returned items; monitors overdue items; manages asset records and user accounts through admin tools.
Borrower / User	Browses assets; submits borrow requests; views borrowed items; requests returns; views history; receives notifications; manages profile and password.

II. SYSTEM REQUIREMENTS

A. Hardware Requirements

Component	Minimum Specification
Processor	Intel Core i3 or equivalent
Memory (RAM)	4 GB
Storage	500 MB free disk space
Internet Connection	Broadband connection at least 1 Mbps
Display	1024 x 768 resolution

B. Software Requirements

Software	Version / Specification
Operating System	Windows 10/11, macOS, or Linux
Web Browser	Google Chrome latest, Mozilla Firefox latest, or Microsoft Edge latest
Web Server	Apache 2.4+ or Nginx
PHP	PHP 7.4 or higher
Database Server	MySQL 5.7+ or MariaDB 10.3+
Hosting Platform	Railway or any PHP-supported web host

III. SYSTEM INSTALLATION GUIDE

The system should be hosted on a live web host so it can be publicly accessible. The following steps describe local setup and deployment preparation.

Step 1: Download the Project Files

Download the ADSSU LAMS project files from the provided repository or source. Extract the ZIP archive to your desired directory.

Name	Date modified	Type	Size
 adssu_lams_borrower	6/12/2026 5:38 PM	File folder	

Step 2: Set Up a Web Server with PHP and MySQL

Install a local server environment such as XAMPP, WAMP, or Laragon. Ensure Apache and MySQL services are running. Alternatively, deploy to Railway using the included configuration files.



XAMPP Control Panel v3.3.0				
Modules				
Service	Module	PID(s)	Port(s)	Actions
☐	Apache	3084 34748	80, 443	Stop Admin Config Logs
☐	MySQL	33316	3306	Stop Admin Config Logs

Step 3: Create the Database

Open phpMyAdmin or a MySQL client. Create a database named `adssu_lams`. Import `database/schema.sql` and optionally `database/seed.sql` for sample data.

File to import:

File may be compressed (gzip, bzip2) or uncompressed.

A compressed file's name must end in `.[format].[compression]`. Example: `.sql.zip`

Browse your computer: (Max: 40MiB)

Choose File

`schema.sql`

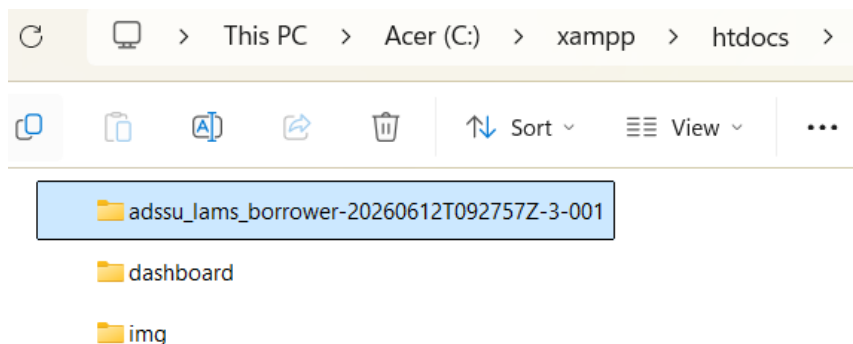
Step 4: Configure Database Connection

The system reads database variables such as `MYSQL_URL`, `MYSQL_ADDON_URI`, `MYSQLHOST`, `MYSQLUSER`, `MYSQLPASSWORD`, `MYSQLDATABASE`, and `MYSQLPORT`. Local defaults may use `127.0.0.1`, `root`, empty password, `adssu_lams`, and port `3306`.

```
MYSQL_DATABASE="railway"
MYSQL_PUBLIC_URL="mysql://$${MYSQLUSER}}:$${MYSQL_ROOT_PASSWORD}}@$${RAILWAY_TCP_PROXY_DOMAIN}}:$${RAILWAY_TCP_PROXY_PORT}}/$${MYSQL_DATABASE}}"
MYSQL_ROOT_PASSWORD="CERHduFTpbdxJSaXfsHcDdmaozkGHQXV"
MYSQL_URL="mysql://$${MYSQLUSER}}:$${MYSQL_ROOT_PASSWORD}}@$${RAILWAY_PRIVATE_DOMAIN}}:3306/$${MYSQL_DATABASE}}"
MYSQLDATABASE="$${MYSQL_DATABASE}}"
MYSQLHOST="$${RAILWAY_PRIVATE_DOMAIN}}"
MYSQLPASSWORD="$${MYSQL_ROOT_PASSWORD}}"
MYSQLPORT="3306"
MYSQLUSER="root"
```

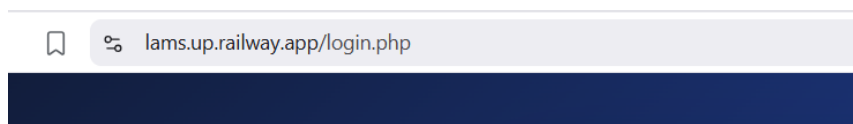

Step 5: Place Files in Web Directory

Copy the `adssu_lams_borrower` folder to the server document root such as `C:\xampp\htdocs\adssu_lams_borrower`, `C:\wamp\www\adssu_lams_borrower`, or `C:\laragon\www\adssu_lams_borrower`.



Step 6: Access the System

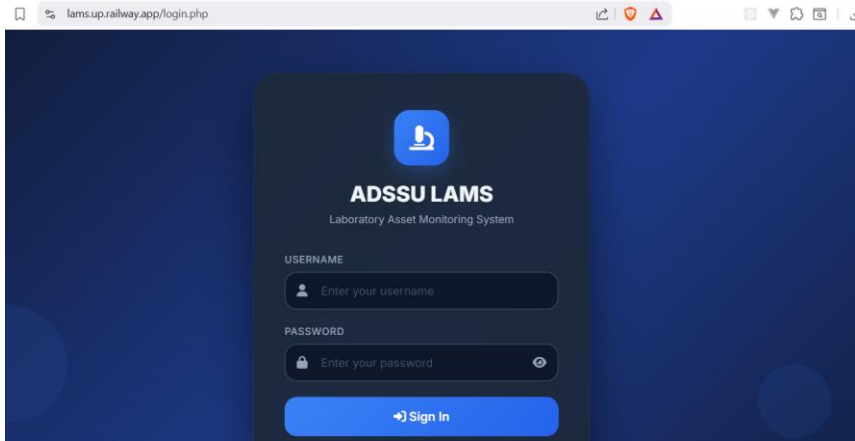
Open a browser and navigate to the local URL or the deployed Railway URL. The system will redirect to the login page.



IV. LOGIN AND USER ACCESS

A. Login Page

The login page contains a Username field, Password field, Sign In button, and Create New Account link.



Login Procedure:

1. Open the system URL in a browser.
2. Enter your username.
3. Enter your password.
4. Click Sign In.
5. After successful login, the user is redirected to the Admin Dashboard or Borrower Dashboard.

B. User Roles

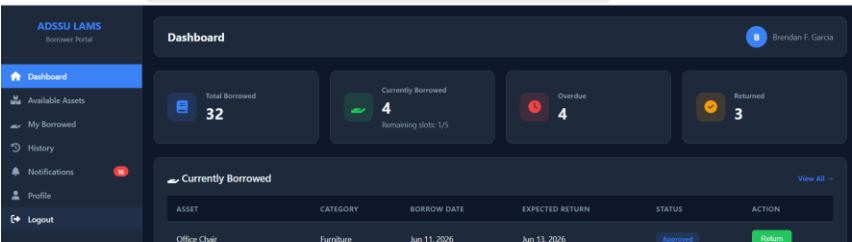
Role	Permissions and Capabilities
Administrator	View admin dashboard, review pending borrow requests, approve or deny requests, confirm returns, view overdue items, and manage records.
Borrower / User	View dashboard, browse assets, search/filter assets, submit borrow requests, view borrowed items, request returns, cancel pending requests, view history, receive notifications, and update profile.

V. SYSTEM FEATURES AND OPERATIONS

A. Borrower Dashboard

Purpose: Provides an overview of borrower account activity and quick access to currently borrowed items.

Procedure: Log in with a borrower account. View Total Borrowed, Currently Borrowed, Overdue, and Returned statistics. Review recent borrowed items and click Browse Assets or View All when needed.

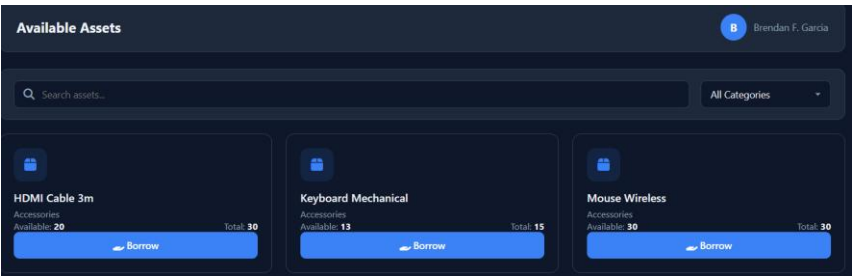


Expected Output: Dashboard with real-time statistics and list of currently borrowed items.

B. Available Assets

Purpose: Allows borrowers to view all laboratory assets available for borrowing.

Procedure: Click Available Assets in the sidebar. Browse asset cards, use the Search bar, and select a Category filter.



Expected Output: Grid of asset cards matching the search or filter criteria.

C. Borrow an Asset

Purpose: Allows a borrower to submit a request for a laboratory asset.

Procedure: Open Available Assets, click Borrow, enter quantity, expected return date, and purpose, then click Submit Request.

Request to Borrow

Asset Name

HDMI Cable 3m

Quantity (Max: 20)

1

Expected Return Date

mm/dd/yyyy

Purpose

Why do you need this asset?

Submit Request

Expected Output: A success notification appears and the item appears in My Borrowed with Pending Approval status.

D. My Borrowed Items

Purpose: Displays all active borrowed items and available actions.

Procedure: Click My Borrowed. Review asset details, dates, quantity, status, and actions. Use Request Return, Cancel Request, Print, or Export to Excel when applicable.

My Borrowed Items

Brendan F. Garcia

My Borrowed Items

Print

Export to Excel

ASSET NAME	CATEGORY	BORROW DATE	EXPECTED RETURN	QUANTITY	STATUS	ACTION
System Unit	Electronics	Jul 15, 2023	Jul 22, 2023	1	Return Requested	Waiting for approval
Tablet	Accessories	Jan 18, 2024	Jan 25, 2024	2	Return Requested	Waiting for approval
Monitor	Electronics	Aug 10, 2024	Aug 17, 2024	1	Return Requested	Waiting for approval

Expected Output: Table of active borrowed items with status badges and action buttons.

E. Borrow History

Purpose: Displays the complete history of borrow requests made by the user.

Procedure: Click History. Review asset details, borrow dates, return dates, quantity, and status. Use Print or Export to Excel when needed.

Borrow History						
ASSET NAME	CATEGORY	BORROW DATE	EXPECTED RETURN	ACTUAL RETURN	QUANTITY	STATUS
Office Chair	Furniture	Jun 11, 2026	Jun 13, 2026	-	5	Approved
Keyboard Mechanical	Accessories	Jun 11, 2026	Jun 30, 2026	-	2	Approved
HDMI Cable 3m	Accessories	Jun 09, 2026	Jun 10, 2026	-	1	Denied
UTP Cable	Accessories	Jun 08, 2026	Jun 09, 2026	-	1	Overdue

Expected Output: Complete chronological list of borrow transactions.

F. Notifications

Purpose: Displays real-time status updates for borrow requests and system alerts.

Procedure: Click Notifications. Review approval, denial, return confirmation, and overdue alerts. Mark notifications as read or delete them.

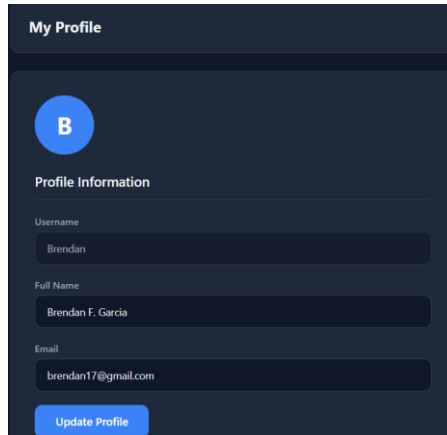
All Notifications		Mark all as read
●	Your borrow request for Office Chair has been approved. You can now pick up the item.	June 11, 2026 12:41 AM
●	Your borrow request for Office Chair has been submitted. Waiting for approval.	June 11, 2026 12:40 AM
▲	The item Digital multimeter is now overdue. Please return it as soon as possible.	June 11, 2026 12:36 AM

Expected Output: List of notifications with icons, timestamps, and read/unread status.

G. Profile Management

Purpose: Allows users to update profile information and change password.

Procedure: Open Profile, update Full Name or Email, and click Update Profile. To change password, enter current password, new password, and confirm password.



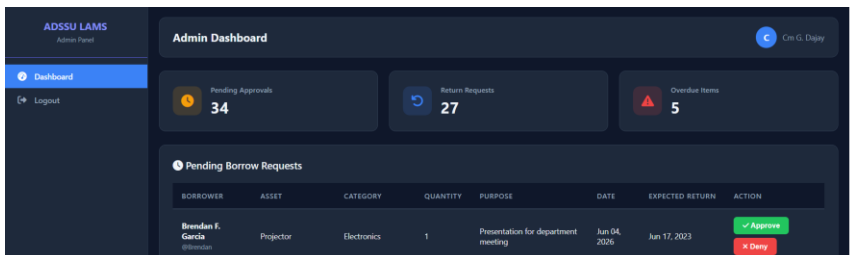
The 'My Profile' form is displayed on a dark blue background. It features a circular profile picture placeholder with the letter 'B'. Below this, the 'Profile Information' section contains three input fields: 'Username' with the value 'Brendan', 'Full Name' with the value 'Brendan F. Garcia', and 'Email' with the value 'brendan17@gmail.com'. A blue 'Update Profile' button is located at the bottom of the form.

Expected Output: Profile information or password is updated with a success message.

H. Admin Dashboard

Purpose: Provides tools for managing borrow requests and monitoring system activity.

Procedure: Log in as administrator. View Pending Approvals, Return Requests, and Overdue Items. Use Approve, Deny, and Confirm Return buttons.



The Admin Dashboard is shown with a dark blue theme. The top navigation bar includes the 'ADSSU LAMS' logo and a user profile icon for 'Cris G. Dapay'. The main content area is divided into three sections: 'Pending Approvals' with a count of 34, 'Return Requests' with a count of 27, and 'Overdue Items' with a count of 5. Below these is a table for 'Pending Borrow Requests'.

BORROWER	ASSET	CATEGORY	QUANTITY	PURPOSE	DATE	EXPECTED RETURN	ACTION
Brendan F. Garcia (brendan17)	Projector	Electronics	1	Presentation for department meeting	Jun 04, 2026	Jun 17, 2023	<button>Approve</button> <button>Deny</button>

Expected Output: Admin dashboard with action tables for managing the borrowing workflow.

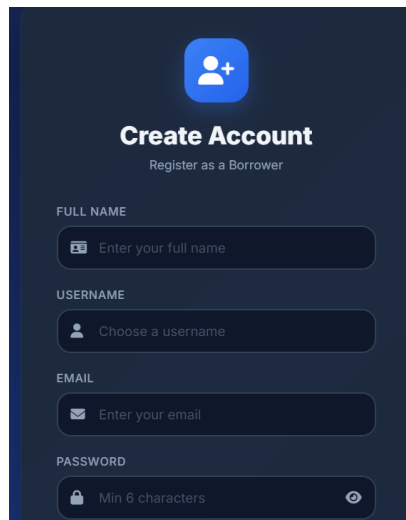
VI. CRUD OPERATIONS GUIDE

A. Creating Records (CREATE)

Creating records adds new data to the system, such as user accounts and borrow requests.

Create a User Account:

6. On the Login page, click Create New Account.
7. Enter full name, username, email, password, and confirm password.
8. Click Register.
9. Sign in using the created account.

A dark-themed mobile app interface for creating a new account. At the top is a blue circular icon with a white person silhouette and a plus sign. Below it, the text 'Create Account' is centered in white, followed by 'Register as a Borrower' in a smaller font. The form consists of four vertically stacked input fields, each with a label and an icon: 'FULL NAME' with a document icon, 'USERNAME' with a person icon, 'EMAIL' with an envelope icon, and 'PASSWORD' with a lock icon. The password field includes a hint 'Min 6 characters' and a toggle eye icon on the right.

Create a Borrow Request:

10. Go to Available Assets.
11. Click Borrow on an asset.
12. Enter quantity, expected return date, and purpose.
13. Click Submit Request.

B. Viewing Records (READ)

Records can be viewed through Available Assets, My Borrowed, History, and Admin Dashboard.

Pending Borrow Requests							
BORROWER	ASSET	CATEGORY	QUANTITY	PURPOSE	DATE	EXPECTED RETURN	ACTION
Brendan F. Garcia @Brendan	Projector	Electronics	1	Presentation for department meeting	Jun 04, 2026	Jun 17, 2023	✓ Approve ✗ Deny
Cm G. Dajay @Cm	IJ 45	Accessories	8	Network maintenance	Jun 04, 2026	Apr 29, 2026	✓ Approve ✗ Deny
Brendan F. Garcia @Brendan	Projector	Electronics	1	Research presentation	Jun 04, 2026	Mar 17, 2026	✓ Approve ✗ Deny

C. Updating Records (UPDATE)

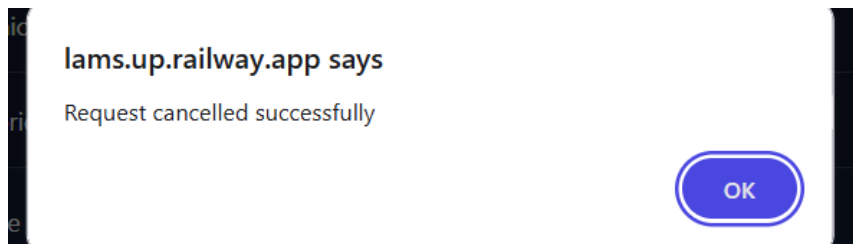
Users can update profile information and change password. Administrators can approve or deny requests and confirm returns.

- Go to Profile to update full name or email.
- Use Change Password to update password.
- Admin users can approve, deny, or confirm returns from the Admin Dashboard.

D. Deleting Records (DELETE)

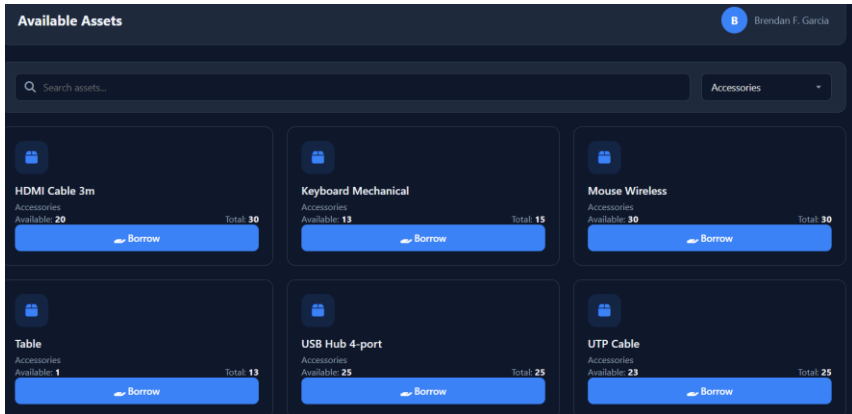
Deletion-related actions include cancelling pending borrow requests and deleting notifications.

- Go to My Borrowed and find a pending request.
- Click Cancel Request and confirm the action.
- Go to Notifications and click the trash icon to delete a notification.



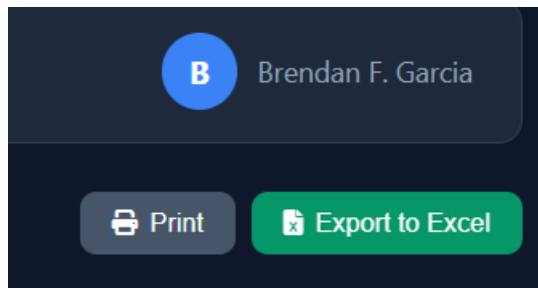
VII. SEARCHING AND FILTERING RECORDS

The system allows users to search assets by name or category. The Category dropdown can display only assets under a selected category. Search and category filters may be used together to narrow results.



VIII. REPORT GENERATION

Users can print records or export them for documentation. In My Borrowed or History, click Print to open a print-ready table. Click Export to Excel to download an XLS or CSV file.



IX. ERROR HANDLING AND TROUBLESHOOTING

Issue	Possible Cause	Solution
Unable to Login	Incorrect username or password	Verify credentials or create a new account.
Database Connection Error	MySQL service not running	Start MySQL service in XAMPP/WAMP control panel.
Page Not Loading	Internet connection problem	Check network connection and try again.
Username already exists	Username is already taken	Choose a different username.
Passwords do not match	Password mismatch	Ensure both password fields are identical.
Cannot borrow more items	Maximum borrow limit reached	Return some items before borrowing new ones.
Could not approve request	Request already handled	Refresh the admin dashboard.
Session expired	Inactivity for more than 30 minutes	Log in again.

X. FREQUENTLY ASKED QUESTIONS (FAQ)

1. How do I reset my password?

Go to Profile and use the Change Password section.

2. How do I edit my profile information?

Go to Profile, modify your Full Name or Email, then click Update Profile.

3. How many items can I borrow at once?

You can borrow a maximum of 5 items at any given time.

4. How long can I borrow an item?

The maximum borrowing period is 14 days.

5. How do I know if my request was approved?

You will receive a notification in the Notifications page.

6. What happens if I do not return an item on time?

The system automatically marks the item as Overdue.

7. Can I cancel a borrow request?

Yes, while the request status is Pending Approval.

8. How do I return a borrowed item?

Go to My Borrowed, click Request Return, and wait for admin confirmation.

XI. SYSTEM MAINTENANCE AND BACKUP

Database Backup

20. Open phpMyAdmin and select the adssu_lams database.
21. Click the Export tab.
22. Choose Quick export method in SQL format.
23. Click Go to download the backup file.

```
mysqldump -u root -p adssu_lams > adssu_lams_backup.sql
```

Database Restoration

24. Open phpMyAdmin and select the adssu_lams database.
25. Click the Import tab.
26. Choose the backup SQL file.
27. Click Go to restore the database.

```
mysql -u root -p adssu_lams < adssu_lams_backup.sql
```

User Account Maintenance and System Updates

- Admin accounts are managed through the admin tools or database.
- Borrower accounts may self-register through the registration page.
- For updates, replace project files, run schema changes if needed, and test all features after update.

XII. APPENDICES

Appendix A: Sample User Accounts

Username	Password	Role	Full Name
Cm	cm12345	Admin	CM G. DAJAY
Brendan	brendan12345	User	Brendan F. Garcia

Appendix B: Database Schema

- users - stores user accounts
- assets - stores laboratory assets
- borrow_history - stores borrow transactions
- notifications - stores user notifications

Appendix C: Sample Assets

Asset Name	Category	Total Qty	Available Qty
Printer	Electronics	13	11
Router	Electronics	20	20
Projector	Electronics	12	10
Monitor	Electronics	50	49
System Unit	Electronics	7	7
UTP Cable	Accessories	25	24
RJ 45	Accessories	25	24
Table	Accessories	13	13
CPU	Components	24	24
Digital Multimeter	Measuring Instrument	5	5

Appendix D: Borrow Status Flow

Pending -> Approved -> Overdue

Pending -> Denied

Approved -> Return Requested -> Returned

Appendix E: References

- PHP Documentation
- MySQL Documentation
- Railway Platform
- Font Awesome Icons
- Agusan del Sur State University



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User Manual

This manual serves as a guide for installing, accessing, navigating, and using the ADSSU Laboratory Asset Monitoring System (LAMS).



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